

Abstract

Experiments have shown that the association of multi-agent systems and role-playing games (RPG) was relevant to tackle the complexity of dynamic social systems sharing common resources. Now, some are used in participatory processes as group decision support tools to favor exchanges of information between stakeholders. In a RPG, stakeholders are placed in a virtual world where roles and rules are characterized. In this approach, do they follow the elements given by the game or do they use parts of their own reality?

As part of an effort to understand if these models can be useful and what consequences do their use have on social equilibrium, the present study is focused on the link between play and reality in negotiation processes. The research took place in irrigated systems of the Senegal River valley. A methodology is proposed to test what farmers regarded as representative of their reality in the model. The methodology is based on synergy between Bales' method and qualitative interviews to analyze interactions of participants involved in negotiation processes in the reality or in the game.

Results showed that (1) Bales' grid has allowed to compare different photographs of situations in the game, (2) qualitative methods were necessary to grasp relevant data, (3) the analysis of behaviors and interactions of participants has partly validated roles and rules of the RPG.

The paper ends with a discussion about whether the nature of reality represented is official or unofficial. It concludes that further researches have to be carried out to tackle the link between play and reality and the status of the RPG in a decision process.

Introduction

Experiments, which associate multi-agents systems (MAS) and role-playing games (RPG), have shown synergies to tackle the complexity and the dynamic social systems sharing common resources. A role-playing game was designed to open the black box of a multi-agent model and explain its content to farmers interviewed in irrigated systems of the Senegal River valley. They partly validated the game as a schematic representation of the reality of management in their irrigated schemes. As the research is going on, the game should be used as a group decision support tool with farmers. How this tool is going to be used by farmers? Are they going to accept strictly the rules of the game or do they create new ones? Our main issue is to grasp what is really happening during a session. Our question is **What is the part of social reality in players' interactions to resolve problems occurred in the game?**

Mucchielli defines the role-playing game as a construction of a problematic situation in which people are acting given-parts. He assumes that a RPG is formed by three elements. The **game** is composed with a system of specific rules. It describes also the world where the session is carried out. The **animator**, also called game-master, organises the session. He knows all the rules. **Players** are actors of the game. Each player makes his own role by following the rules (A. Mucchielli, 1983). For us, there are also the **observers** of the game. They do not participate in the game but they are present and they notice crucial data to the analysis of the session.

Both, objective and subjective views are necessary for a complete analysis of the game session. The animator is involved in the game. Even if he is not a player, he is one of the elements of the game. So he may have a subjective view of interactions between players. Because they are outside of the game, observers can register crucial information about what is happening in the session. They give an objective view to analyze the session. Our research is focused on players. They interact, take decision and express viewpoints. From exchanged information, they make a solution to the setting problem. In our research, players belong to the same social community.

The rules partly force the behaviors of actors in the game. A general framework for actors' behaviors is given by the set of rules. But, because rules can not be exhaustive, players have got a certain degree of liberty. Then, the main issue we focused on is to understand whether players really follow the rules of the

game or not. With their degree of liberty, they may choose to create and follow complementary rules or to modify the rules of the game to make it closer to their reality. To tackle this question our research attends to the interactions between players¹. We try to understand if relationships and social norms shared by players overdetermine their interactions in the RPG. For this purpose, we have to observe how people interact in real situation in order to grasp social behaviors and norms shared by these people. To reach our goal we decide to focus on situations where locals have to interact to take a decision, to negotiate to solve a problem occurred in their irrigated scheme.

This paper aims at presenting a methodology to analyze interactions between players committed in a RPG. Players are farmers producing rice in irrigated systems of the Senegal River valley. The RPG was built up as a schematic representation of their real world. The methodology is based on complementary analyses of virtual and real contexts of negotiation. Farmers are embedded in the same social system. Rural credits are essential to produce in their plots. Seasonally they have to face together the problem of sharing insufficient rural credits. The methodology consists in the characterization of situations in which people are interacting to solve *common problems*.

This paper begins with a presentation of the context of the research and a description of the model. The second part describes the methodology used to characterize negotiations in the game and in the reality. The last part puts the methodology back in the context and sheds new light on our main issue.

1. From reality to role-playing game.

We introduce, first, the issue of the local management of rural credits to show that it is (1) a *common problem* essential for the production in the irrigated schemes and (2) the topic of recurrent negotiations between stakeholders. Then, the RPG *Njoobari ilnoowo* is described.

1.1 Choice of rural credits as main topic in the negotiation observed

1.1.1 A constraint : negotiation about the same recurrent topic.

In the real world, we focused on negotiations dealing with rural credits management in irrigated systems because of two reasons. First, the access to rural credits is necessary to produce irrigated systems in the valley of the Senegal river. Secondly, because they are lower than needs, the management of rural credits is seasonally a subject of negotiation. The two next paragraphs show that rural credits are central in the management of these irrigated schemes and are the topic of recurrent problems and negotiation processes.

1.1.2 Rapid historic of the irrigated agriculture in the valley of the Senegal river.

The irrigation in the Senegal river valley increased with the Independence of Senegal obtained in 1960. This political choice was made to support the economic development of the country thanks to the outcomes of agricultural production. Irrigation, mainly financed by international donors, was based on modern hydraulic infrastructures and on the training of farmers. Everything was mechanized. Primary, secondary and tertiary schemes were built to control water even more. After droughts of the 70's, Senegalese government increased its building of tertiary irrigated schemes. The development of the valley evolved differently from a region to another, because of the density of populations. In the delta, sparsely populated, big irrigated perimeters (BIP) were constructed. In the middle and the upper valley, densely populated, village irrigated perimeters smaller than previous ones, intended for small groups of farmers, were built in association with BIP (P. Boivin et al., 1995; G. Diemer et van der Laan E. Ch. W., 1987; B. Crousse et al., 1991). Between 1976 and 1996, irrigated surfaces increased tenfold particularly with the development of private irrigated schemes (L. Liagre, 1997)

Nevertheless, the results were less than funding agencies and Senegalese government expected. The production is very expensive because of the over-mechanization. Thus, farmers became dependent of rural credits to produce in irrigated schemes. Yields stagnate. Infrastructures were deteriorating quickly. The cost of their rehabilitation was weighty. In front of these relative failures associated with economic difficulties, the Senegalese government had to withdraw from agricultural activities. The hydraulic management was transferred to farmer organizations, the financial management to the national bank of rural credits

¹ even if the observers may interfere in players' interactions. But we do not attend to this question.

(CNCAS), and the tenure management to rural communities. The rest of activities were in charge of private enterprises (B. Crousse et al., 1991).

1.1.3 Credits are the topic of recurrent negotiations.

Because of political choices, farmers are now financially dependent from the outside. The cost of mechanized production is so weighty that farmers need credits to produce in irrigated schemes. Usually, farmers' organizations (GEI : Group of economic interest) express their needs to the CNCAS. The bank only gives credits for rice and tomato productions. The credit allocation depends on the repayment of last campaigns credit engaged by the GEI. The bank does not recognize individuals. So, in a GEI, all members are collectively responsible for the credits obtained. They are dependent ones from the others.

Moreover, with the high costs of production in irrigated systems, some farmers have developed strategic behaviors about credit repayment. Now, they allocate credits of the bank to produce different crops in their different plots. This strategy to share risks is partly responsible of low yields in rice production. With low yields, all the credits obtained are not repaid back to the bank. But for the CNCAS, the GEI is still responsible of credit repayment. And, when one of the members does not pay back, all the others have to repay his debt if they want to receive new credits. Otherwise, all the members of the GEI are penalized.

Therefore, credits are in the center of negotiation processes between members of GEIs (farmers), villagers and others members of the social community (living outside the village: migrants, NGO² representatives,...). Exchanges are all the more important because, with the high cost of production, a minimum of farmers producing in the irrigated schemes is necessary.

In each campaign, the problem of credits appears again. And seasonally, villagers have to negotiate to solve the problem. The access and the management of credits are central in the irrigated schemes. They are the support of some social interactions between stakeholders.

1.2 Description of the Njoobaari ilnoowo role-playing game.

This RPG was built in an accompanying approach developed with farmers of irrigated schemes in the valley of the Senegal river (O. Barreteau & al., 2001). This RPG is extracted from the MAS Shadoc (O. Barreteau & F. Bousquet, 2000). It is formally a MAS whose agents are not played by computer entities but by human beings.

1.2.1 Introduction to Njoobaari ilnoowo

The game is played with 10 to 15 players. Players take roles of farmers in an irrigated scheme in the Senegal river valley. They all cultivate only rice in this scheme.

Space is made of two areas, one hidden from the other. In the first one, players are divided in two villages separated by several kilometers and an old dispute. Therefore, there is no communication between villages in this zone. The second area represents the irrigated scheme where players are organized in two farmers' organizations (GEIs). People from both villages can be in the same GEI. The irrigated scheme is drawn on a blackboard. When players are in this zone, they can communicate. Each GEI is in charge of the management of a watercourse. Every player has got only one plot. One pumping station supplies water to every watercourse. An abacus is used to allocate water among the opened plots. The allocation depends on the number of plots of a watercourse opened in the same time and their relative place. Everybody knows this abacus. A factor for the evaporation of water is also applied.

Players follow the rules given in the game. At the beginning, farmers draw three cards at random. These cards written in Pulaar, Wolof³ and French define their basic behavior during the session. Three parameters characterize each player :

- a social status (among a hierarchy of four) that determines the possibilities of exchange of labour or credit (figure 1) ;
- a goal of production in the plot, among a set of three with an increasing level of intensification (figure1).

² Non governmental organization

³ two senegalese languages.

- a rule of repayment of loans among four (figure 1).

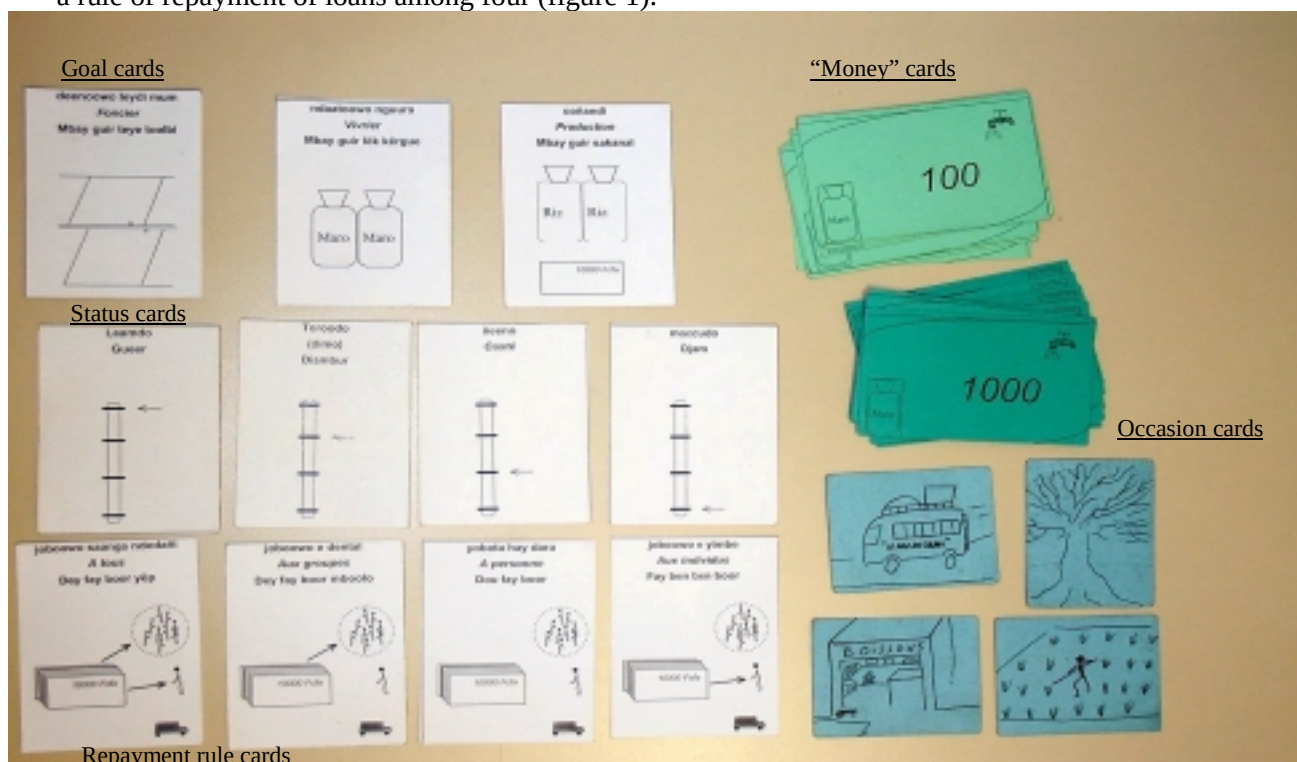


Figure 1 : Njoobaari cards

1.2.2 The RPG progress

The game is the succession of several cropping seasons. During one season, three stages are occurring : search of credits, irrigation/production activities, appraisal of the season (figure 2). On the first season, credits are supposed to be given by the bank to cover all purchases of inputs.

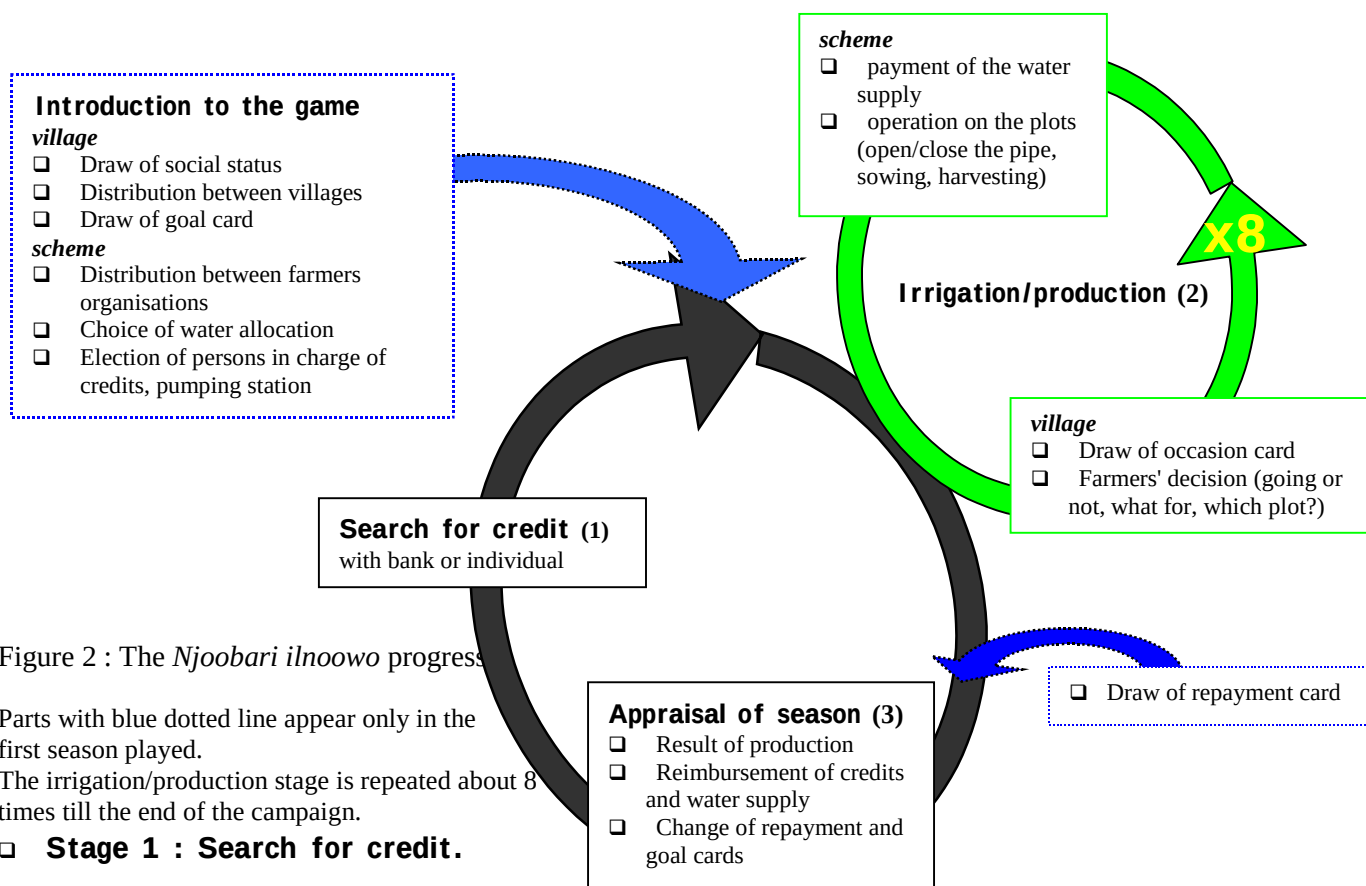


Figure 2 : The Njoobaari ilnoowo progress

Parts with blue dotted line appear only in the first season played.

The irrigation/production stage is repeated about 8 times till the end of the campaign.

- **Stage 1 : Search for credit.**

To cultivate rice in the irrigated scheme, farmers need credits. In this first stage, we suppose they got all the credits they need. They receive different credit according to their goal card. But they will be able to reimburse it during the appraisal stage (stage 3). So at the beginning of the game, this first stage is not played.

□ Stage 2 : Irrigation and production

Before starting the game, each player receives the cards that define his role. They are distributed randomly in the villages and in the farmers' organizations. They choose (i) a chief for each farmers' organization, (ii) a manager of the pumping station and (iii) a person in charge of credits. In the scheme area, the chief decides with his group the allocation rule along his watercourse. So these players have got two roles : a collective one and an individual one. Then the game begins.

In the village area, the players belonging to the first farmers' organization draw randomly an occasion card (figure 1). A combination between the card drawn and their goal card allows them or not to go to the scheme area. In the scheme area, a board is drawn with the position of each player's plot. Each plot is initially dry. Farmers decide to open or close their pipe by following the collective rule of water allocation in their watercourse to irrigate their rice. To cultivate by sowing one of the two varieties of rice (a long one and a short one), they have to pay inputs according to their goal card. They leave this place to return to the village area. Then come the members of the second organization, who are allowed to go to the scheme. At each turn, water level and state of cultivation of each plot is computed according to an abacus (figure 3). Everybody knows this abacus. This is repeated about eight times in a season until the rice is ripe enough to be harvested.

If the pipe of one plot is opened along the watercourse, it receives :	+60
If the pipe of two plots are opened along the watercourse (in the order) :	+40, +20
If the pipe of three plots are opened along the watercourse (in the order) :	+35, +17, +8
If the pipe of four plots are opened along the watercourse (in the order):	+32, +16, +8, +4
If the pipe of five plots are opened along the watercourse (in the order):	+31, +15, +8, +4, +2
If the pipe of six plots are opened along the watercourse (in the order):	+30, +15, +8, +4, +2, +1
(The water level of a plot can not exceed +30)	
For all the plots with a negative water level, the evaporation is :	-3
For all the plots with a positive water level, the evaporation is :	-6

Figure 3 : the abacus

At the end of the season, the manager of the station stops pumping.

□ Stage 3 : Appraisal of the season

When the rice is harvested, farmers sell their products and receive the equivalent in virtual money represented through paddy bags (figure 1). This equivalent depends on their goal card and the number of stress events suffered by the crop. They draw randomly repayment cards among four (figure 1). Each of them describes different behaviors: to pay back to collective creditors (bank and pumping station), individual farmers, both or none. Managers of the bank and of the pumping station sit and wait for farmers' reimbursement. The players in charge of the bank and the pumping station decide the rule of loan among three. Each one can choose a different rule. Then, if players are allowed, they can pay their credits back.

□ Stage 1(2) : Search for new credits.

At this stage, farmers can change their repayment and goal cards. If they need, farmers can get into debt with individual or collective creditors. At the end of this stage, farmers are ready for a new season.

An average session is made of two cropping seasons played in about a half-day. The number of players is between 10 to 15 persons. With less than 10 participants, the game does not reveal the complexity of water

management because they do not have to interact to operate on plots. With more than 15, the animator has many difficulties to manage the game and the observers can not follow the exchanges between players.

2. The methodology : complementary analyses of negotiation processes in the RPG and in social reality.

2.1 Three hypotheses to tackle the link between play and reality.

This methodology was built to give some answers to our main issue : **What is the part of social reality in players' interactions to resolve problems occurred in the game?** In order to shed light on our issue, we broke it up into three interrogations:

- 1- For players, how is the reality represented in the game ?
 - 2- If the players accept this representation of the reality, will they use their real behaviors, norms and relationships in the game ?
 - 3- In that case, will the game be able to reveal these elements to the players and the observers ?
- To tackle these questions three hypotheses should be tested.

<u>Hypothesis 1 : The game is accepted as a schematic representation of reality</u>
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The game was made as a schematic representation of real management of senegalese irrigated systems. But even if it was built up in an accompanying approach, the game is partly an expression of the researchers' point of view. So, the first question was to validate whether farmers accepted this point of view or not. O. Barreteau & al. (2001) have already presented the RPG to them. The interviewed stakeholders have validated partly the proposed representation. Characters presented in the game are caricatures of real roles as farmers have already recognized. Proposed rules are not exhaustive. So, what do they accept exactly: the rules, the types of rôles or both of them? This research tries to go further in the validation of the model. The aim is to analyze behaviors of farmers in situations of action where they have to interact. Negotiation processes in the game and in the reality represent such situations of action. This analysis should be helpful to go beyond the first step of validation already done. Because people in action are dynamic, the analysis of their interactions should grasp full of elements (about norms, behaviors, rules really followed in negotiations,...) not given in the first validation. And these elements are essential to understand how the reality is represented in the game.

“What is the part of the reality involved in the RPG” still remains our main issue. This question is visited here through its aspects of representation.

<u>Hypothesis 2 : The social background of players interferes with roles playing in the game</u>

For J. Huizinga (1951) and R. Caillois (1967), game is out of life. Players are supposed to abandon their social background before playing. Our second hypothesis is opposed to this assumption.

Roles in the game are framed with formal rules. The played characters are not real because their actions are limited by the game. Therefore, actions and behaviors of players are supposed to be under control. Players are members of the same village. They share values, norms, habits and history that determine their behaviors and relationships in the real life. All these elements structure their actions. And this may interfere with their behaviors caricatured in the game. How players use their social background to act in the game? May real social or economic hierarchy be reproduced or changed in the virtuality of the game ?

<u>Hypothesis 3 : The game reveals social relationships between players</u>
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We remind that our main issue is to understand what is the part of social reality in players' interactions to resolve problems occurred in the game. If we accept the second hypothesis, the problem is now to grasp elements of players' social background used in the game. Do the game be helpful to reveal these elements to observers?

The RPG speeds time up and compresses real space. Two seasons, representing a year in the reality, are played in a half-day. Players can go from the village to the irrigated scheme in few minutes whereas in reality it can be several kilometers apart. Therefore, in one place different stakeholders may be represented and express, defend and justify their viewpoints. In this compressed atmosphere, may social relationships become perceptible, obvious, or understandable to the researcher? Or are players so embedded within their common social background that they do not need to express them in the game?

In this paper, we focus on the first hypothesis. Thus, the methodology should be helpful to tackle through villagers' play how the reality is represented in the game by

- 1- recording the elements to describe and understand individual or collective players' behaviors in the game
- 2- grasping behaviors that belong to the rules of the game *sensus stricto* and those which do not.

To reach these aims, to register information in the most objective way, to return to information exchanged in negotiation meetings if necessary, we decide to use a camcorder.

The methodology should provide sources of observed deviancies. Following the hypothesis done, the methodology explores individual or collective farmers' behaviors in real processes of negotiation on rural credits. Therefore, **the methodology is based on a synergy between real and virtual contexts, the one highlighting the other and vice-versa.**

Test this hypothesis implies to register all relevant information in the game and in the social reality to piece together the stages of negotiation processes. The methodology is developed in order to characterize negotiation process in the game and in the social reality. Thus, the methodology is :

- 1- to set up a system to register information occurring in the role-playing game, to analyze them, and then come to players with specific questions to better understand their behavior and actions during the game.
- 2- to identify stages, stakes and stakeholders in real negotiation process on rural credits.

2.2 Recording material to pick up information

2.2.1 In the role-playing game

A sheet was created to record all physical operations on the irrigated scheme. A facilitator fills in this sheet. All cards (occasion, repayment, goal, status) are noticed by each player on a personal sheet. On each turn, they indicate actions they want to do before. After going to the scheme, they indicate actions they have really done. The sheet for players is very simplified. Symbols were used to allow illiterate players to record information too (figure 4). Therefore, all formal actions are registered.

The camcorder is used to record negotiation in the irrigated scheme area. The camcorder is located there till the end of the campaign. During the reimbursement and the search for credits phases, the camera takes place in the village area.

Date :

Doggol wad be gollal ngal

Rogere.....

Wuro.....

Innde e yettoode:

Hinde:

Faandaare:

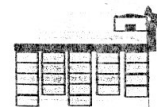
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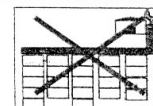


Yahde to kolongal

golle

alkule

	a : demal	a1 a2
	b : ilnude	b1 b2
	c : ilnaani	c1 c2
	d : sarde	d1 d2
	e : teretmaa	e1 e2
	f : wadde engeree	f1 f2
	g : soñde	g1 g2
	h : godnum	h



Waasde yahde to kolongal

X

Figure 4: Players' sheets

2.2.2 In the real negotiation process

In real negotiation, the video is not easy to use for two reasons. First, because the negotiation process is not localized in space and time. Second, because stakeholders may refuse the presence of the video. Therefore, the record of information is done by writing minutes of meeting on credits and sociological interviews. The aim of these interviews is to construct *a posteriori* the history of the negotiation process about credits on the last or actual campaign. The cross-checking of information given by different sources is realized. Stages, stakes, stakeholders and simple farmers are identified and interviewed about their place in the process, their actions, their viewpoint about the solution found.

Video recording is used because many interactions occur between more than two players. They are moving in the space, expressing their actions and reactions with their body. So, a lot of information appear in the game and it is difficult for an observer to record them all. To increase the number of observers could have been an idea, but the number of information conveyed and the limit of observer have also directed the choice to a video recording system. The problem is the importance of the information. This main difficulty imposes to think of the methods used to classified information helpful in the analysis.

2.3 Analysis of negotiation processes

Negotiation processes in the game and in the reality are analyzed through two aspects : the interactions of actors and the topics developed in the discussion.

2.3.1 Analysis of interactions between actors with the Bales' grid (1950)

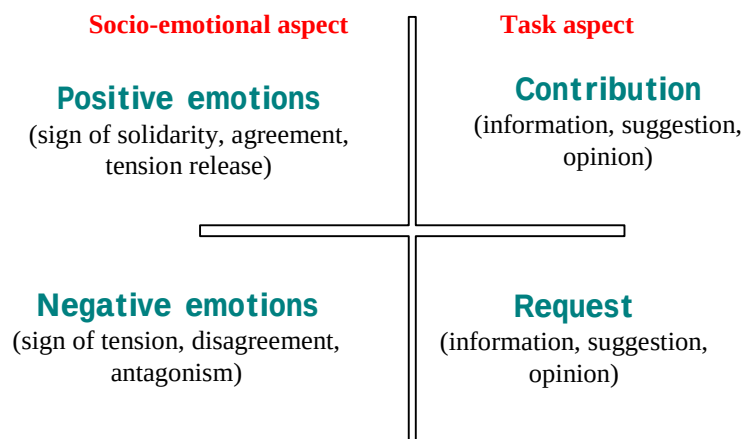
□ Bales' method

Theoretical foundations

Two theoretical trends in social psychology analyze the behaviors of persons involved in a negotiation. The first one describes what people are doing : this is the behavioral approach of decision making. The second one focalized its analysis on relationships between actors : this is a type of interactionist approach (Bazerman M.H. et al., 2000). Bales belongs to the second one.

He tried to make empirically a detailed list of behaviors to describe social interactions within small groups. He had built a list of 12 categories. Six belong to the "task" and six else to "socio-emotional" aspects. He proposed an analysis of the general functioning of "social actions" that characterizes small group. Bales considered that small groups go successively from task position to social-emotional problems. First, members of a group focus on instrumental aspects to reach the goal they are organized to. But their activity creates socio-emotional reactions which become more and more important in the group. At a certain point, members of the group decide to concentrate and resolve socio-emotional difficulties to go further in their instrumental task. They defuse these socio-emotional problems and then instrumental troubles appear again, and so on till they reach their final goal. The members of a group should communicate, should be able to make an assessment of the situation and then control the features to solve the presented problem (R.F. Bales, 1950).

Figure 5 : Simplified Bales' grid



The method

Considering two individuals X or Y members of a group. X says a sentence P1 to Y. Y answers to X by saying P2. An **interaction** is defined as a sequence of sentences said by one person. Here, the interactions

are coded with the four main categories and numbered. Each actor's speech is classified in one of the categories. In the figure 6, Actor 2 welcomes players (interaction 1, noticed in positive emotion). Then, Actor 3 presents to the group the goals of the meeting (interaction 2, contribution). With the sums, some graphics showing the structure of the discussion analyzed can be made.

Participants	CATEGORIES					
	Positive emotions	Negative emotions	Contribution	Request	Hubbub	Total
1		7	6		10	2
2	1	8		4-14		4
3			2			1
4				3-13	10	2
5	11		5-12			3
6		9			10	1
...						0
Total	2	3	4	4		

Figure 6 : Coding of interactions

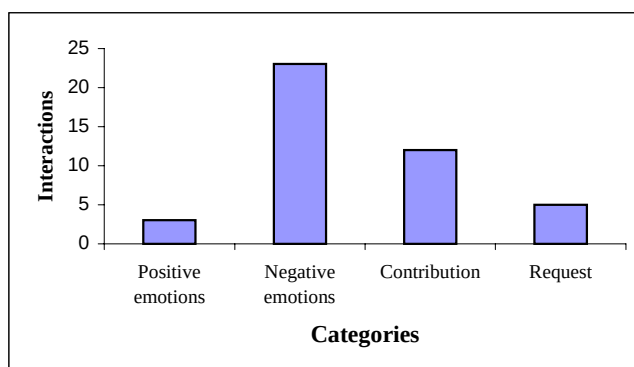


Figure 7 : Interactions=f(categories)

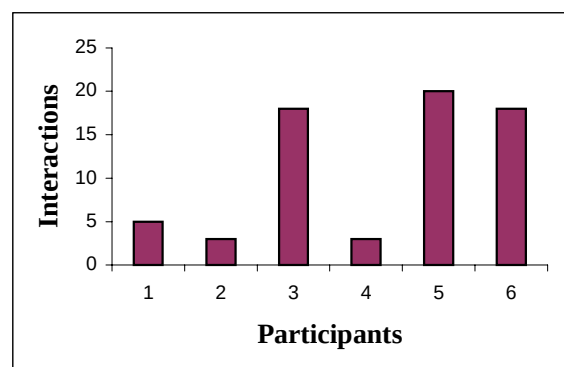


Figure 8 : Interactions=f(participants)

The figure 7 gives information on the nature of interactions appeared during the discussion. For example, if the categories "negative emotions" and "contribution" are more important than the others, this show that the discussion was tense and all the actors have express their viewpoint without listen to the others. The figure 8 gives information about who is the more active in the discussion.

Interest and limit of the grid

A video recording is necessary before applying the Bales' method. All the interactions registered are translated from Pulaar to French and then transcribed. Because of translation, we decided not to use the twelve categories but only the four principle classes. Here, it is essential to recall what we expect with this method : **a characterization of interactions between participants involved in negotiation process in the game or in the reality.** We are seeking a way to analyze relationships between participants in order to understand how the reality is represented in the game.

The first interest of the Bales' method is to give information about the dynamic of the group : positions of interacting people and the evolution of positions as the negotiation is progressing. But it is mainly focused on expressed communications. And all the relations between members of a group are not conveyed through verbal medium.

The main interest of the grid is to take picture of different situations of interactions. These photographs can be helpful to compare a situation to another. The grid should give us a clue to characterize social interactions between actors. With this grid, because its parameters are limited (even if it can not be

exhaustive enough to understand all the factors determining actors' behaviors) the comparison of different negotiation meetings can be done on two aspects: the traffic of verbal information, the actors' positions in the discussion.

So the methodology can be useful to follow what are the (virtual or real) information exchanged and how it is done. It gives a quantitative view to our analysis.

But some limits must be underlined.

- The coding of interactions is subjective. However, we consider that the subjectivity of a same observer is little variable.
- One parameter determines the importance of actors in the discussion : his talkative character. Nevertheless, a talkative person in a group is not necessary the most influent in the solution found to solve the problem faced.
- Social networks of actors not based on communication are missing in this method.

2.3.2 Qualitative analysis of negotiation processes

The last two points are very important in our research. To fill these two limits, we decide to complete Bales' grid with qualitative methods.

First, we pay attention to the topics of the discussions in the negotiation. We consider that during the whole process of negotiation, people are not involved in the same way in all the themes of a discussion. So, we have to notice data concerning : topics and stakes of discussion, who takes part in the discussion, what are his arguments, does he succeed or fail to convince other members, how does he manage, who are the members sharing or facing his point of view.

Secondly, after the negotiation meetings or at the end of the game, we make (individual or collective) debriefings with participants to analyze remarkable interactions the observer has noticed. During debriefings, the relationships between participants can be pointed out. Sociological interviews are done in order to complete the video-registered data.

With this set of methods to analyze negotiations, the interactions between participants and the themes developed in the discussions are investigated to grasp processes of negotiation in the game and in the social reality. The synergy between these analyses is necessary to understand what part of reality is represented and involved in the playing of actors. But how are these analyses combined in real and virtual negotiations?

3. Put the methodology back in its context.

The methodology was developed in Wuro Madiu village, located in Podor department, in the middle of Senegal River valley. Wuro Madiu was chosen because the accompanying approach to build the MAS and the RPG was partly set up there.

The present methodology was adapted to the two contexts of analysis. It was wholly developed in a virtual world during game sessions. In the real world of irrigated systems, it was partly developed during negotiation about the management of rural credits. The investigation of real negotiation processes is made to highlight observations and analyses of negotiation processes in the RPG. On the other hand, the analyses of the distance between played roles and expected roles (with the rules given) provides information on real social parameters that influence the negotiation in the game. The two contexts are complementary to the research since one is helpful to understand the other and *vice versa*. Put the methodology back in its context is also necessary to obtain a dynamic analysis.

3.1 In real negotiation on rural credits

The analysis of real negotiation processes depended, first, on the occurrence of a meeting on credits during our presence, and then, on the acceptance of a video by farmers. The question about the influence of a camcorder on the discussions between villagers is not what it is asked here. As Duranti we consider that our presence in the village has already influenced the social life. It is just a case of the observer-participant paradox (A. Duranti, 2002).

We had imagined how the methodology could be developed in different cases

- 1- There is a meeting and the use of a camcorder is accepted. All the analyses are done (as in the RPG). Sociological interviews are realized to piece together the historical elements of the negotiation process by multiplying viewpoints. These interviews are focused on stakeholders present when the final decision is taken, persons who played a part during the process and farmers affected by the results of negotiation. Feedback interviews are essential to this analysis.
- 2- There is a meeting but the camcorder is not accepted. If the presence of an observer is allowed, thematic and sociological feedback interviews to analyze negotiation process.
- 3- There is no meeting. Then we use only sociological feedback interviews.

When we tried to develop our methodology in Wuro Madiu we were confronted with this third case. That's why Bale's method was not used in real context. We have made sociological feedback interviews to analyze process of negotiation on rural credit during the past raining season. The bank (CNCAS) gave credits according to their reimbursement rate, because farmers had not repaid all their rural credits back. As we have already explained, the production in irrigated systems need credits. There were two ways villagers evoked to allocate the funds obtained. In order to maintain social cohesion, some preferred to find a way to distribute money to all farmers. Farmers would be able to produce but only in a part of their plot, part function of their reimbursement rate. On the contrary, others (called the 100% group) had made sacrifices to repay all their credits back. These people wanted to keep the credit to cultivate their whole plot. Several formal or informal negotiation meetings were made to solve the problem. The figure 9 shows how some farmers called a facilitator (AB) to help them. He met men of power (the head of the village, the imam: religious leader) in the village and succeeded to convince them to join his point of view. They legitimated him to form a small group to think about a solution to the problem. After several discussions in comity, the first proposition was exposed to all the farmers. The majority voted for the proposition.

Legend:

Galle = concession (in pulaar)

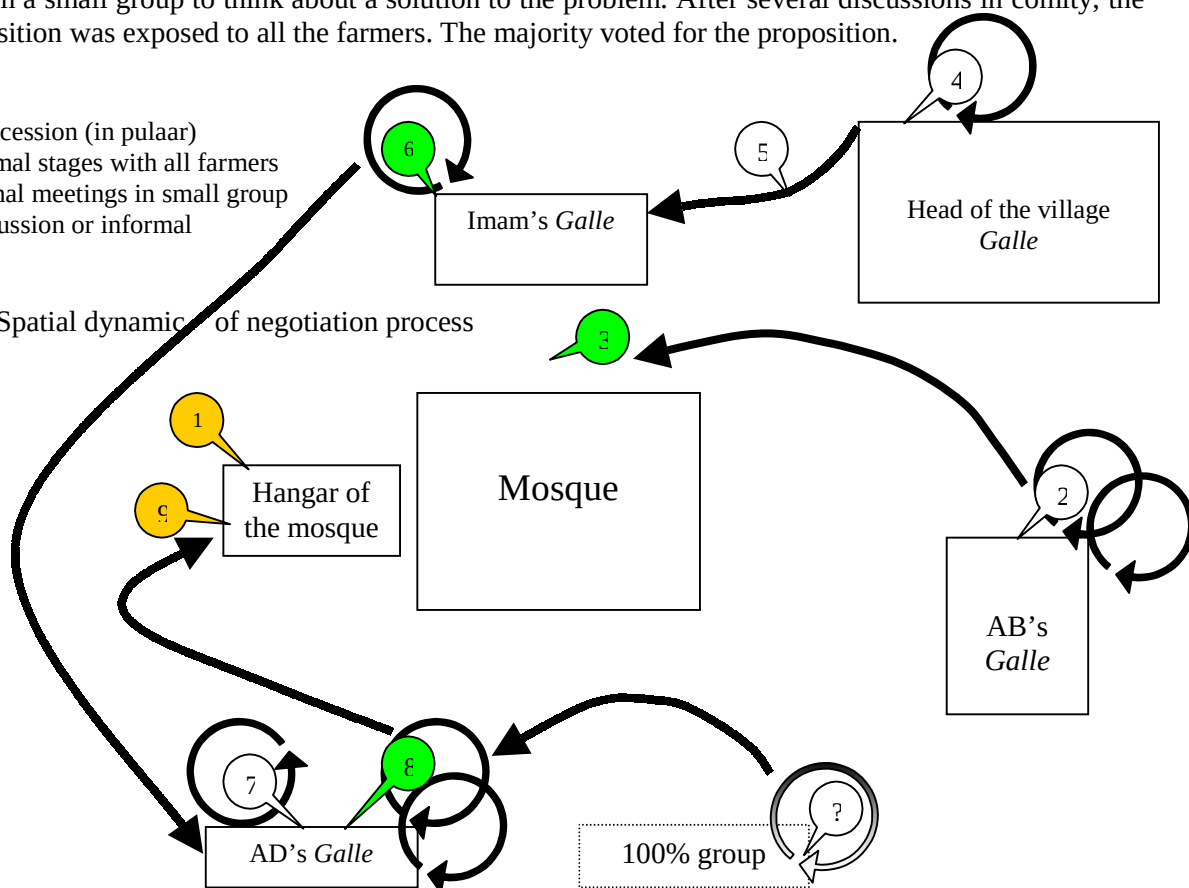
yellow: formal stages with all farmers

green : formal meetings in small group

white : discussion or informal meetings

meetings

Figure 9 : Spatial dynamic of negotiation process



3.2 In the role-playing game

The game was played twice with 12 and 13⁴ farmers. Wuro Madiu is a Muslim village. An announcement was made at the end of the Friday prayer in the village mosque to all villagers to invite them to the

⁴ So, we are just in the conditions of small groups and the methodology proposed can be applied.

presentation of the research. During this meeting, names of all persons interested in the game were written down. In both sessions, two campaigns were played in a day. The camcorder was located in the scheme area. After the game, a debriefing was carried out with players to make them expressed their opinion of the game and explained to us some behaviors we were not expected to. During this debriefing, people interact directly with the group, justify their actions and opinions and reveal clues to our understanding of the negotiation process. Video data were translated and transcribed. Then, a second debriefing was done individually with players whom behavior was remarkable. The whole methodology was applied to analyze players' interactions.

With Bales' graphs (figure 10) we made photographs of the interactions occurred during the game. We are able to compare one tour to another or one campaign to another one. The biggest part of the communications between players was focused on the course of the game. So a lot of interactions coded belong to the domain of the task : players gave information about what they wanted to do or about actions of the others, they expressed their opinion,... In the second campaign of the second group a problem appeared in the 2nd tour about the distribution of water and tensions reached their maximum in the 4th tour. As we felt during the game, communications between people have left the task domain to belong to the socio-emotional one. During these tours, the number of interactions was more important. In fact, some people wanted to keep on playing whereas others asked to treat the problem even by stopping the game course. The graph shows those elements. The comparison of the two campaigns of the first group shows also that interactions were less numerous in the second one than in the previous. We consider that Bales' graph gave us a tangible example of the players' learning of the game.

As we have applied the Bales' method we are now able to go further in its criticism.

First, it is a very heavy method because the transcription and the coding of communications takes a lot of time. To make the methodology lighter, it is necessary to identify remarkable sequences where Bales' grid could be apply. Second, when the communications are numerous (more than 20), graphic of the dynamic of interactions between participants is unreadable. Third, Bales' method is focused on communications and moreover it constraints the way to grasp information. Thus, it gives partial data about the behaviors of participants and about their interactions. So to reach our goal of a description of interactions, the Bales' grid must be used with others methods. Here we join some criticisms of the clinician approach. Clinician psychologists feel unease with Bales' method because they reject grid made *a priori* to observe phenomena. They consider that because they are supposed to grasp systematically some phenomena, grids can miss those unpredictable (Anzieu D. & Martin J.-Y., 1997). But in another hand, this formatting is interesting because the observer must be rigorous in the coding. Because he must stay in the frame, this framing is helpful to make comparison of different sequences of discussion.

3.3 In the research issue

Now, we want to put the methodology back to our first hypothesis : what part of reality is represented in the game ? Here we analyze qualitative data coming from the different interviews made during the debriefing of the game, the Bales' method and thematic analyses. It is important to make a difference between what actors say about the reality represented in the game and what we interpreted as involvement of the reality in the game.

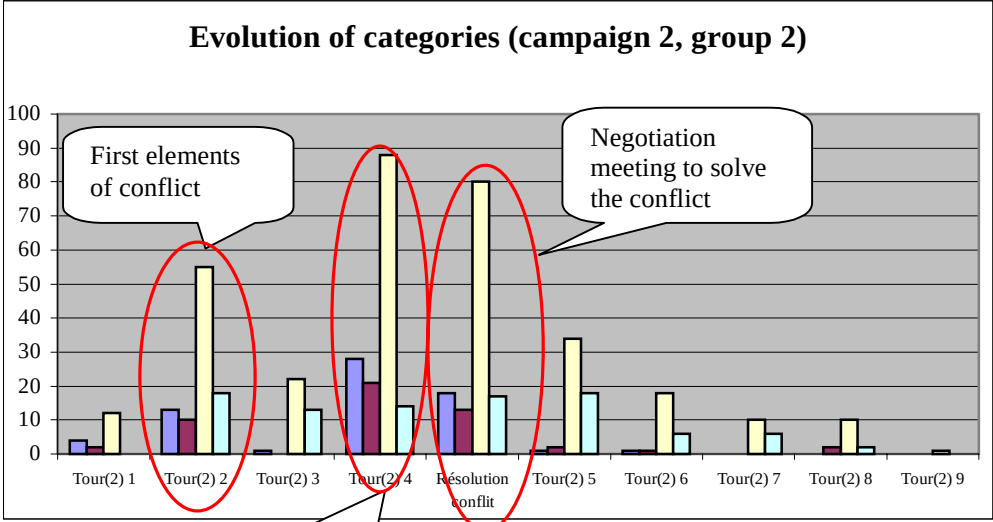
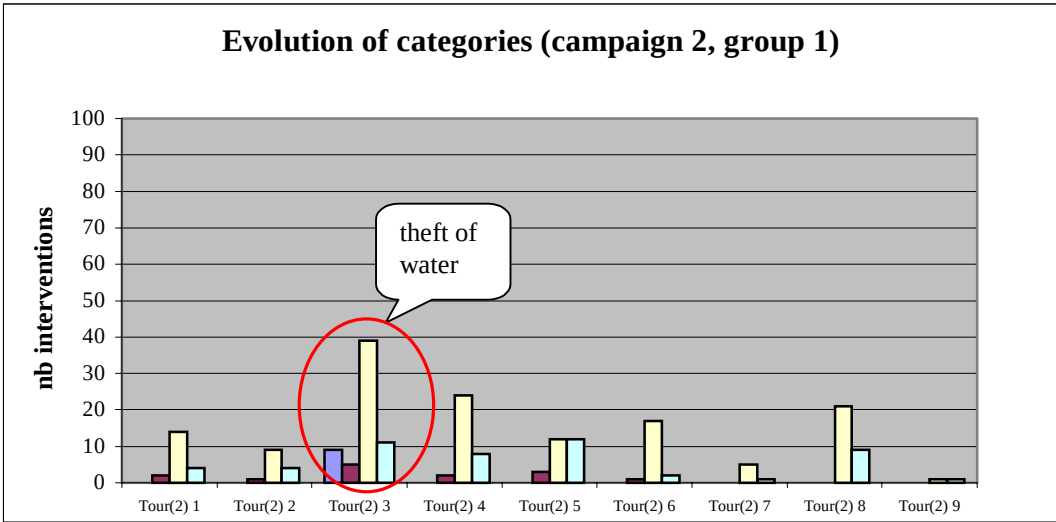
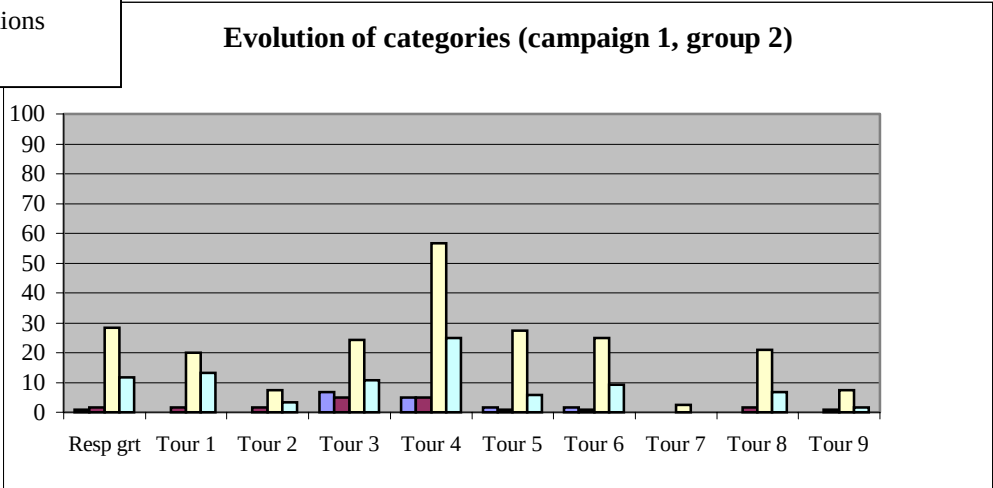
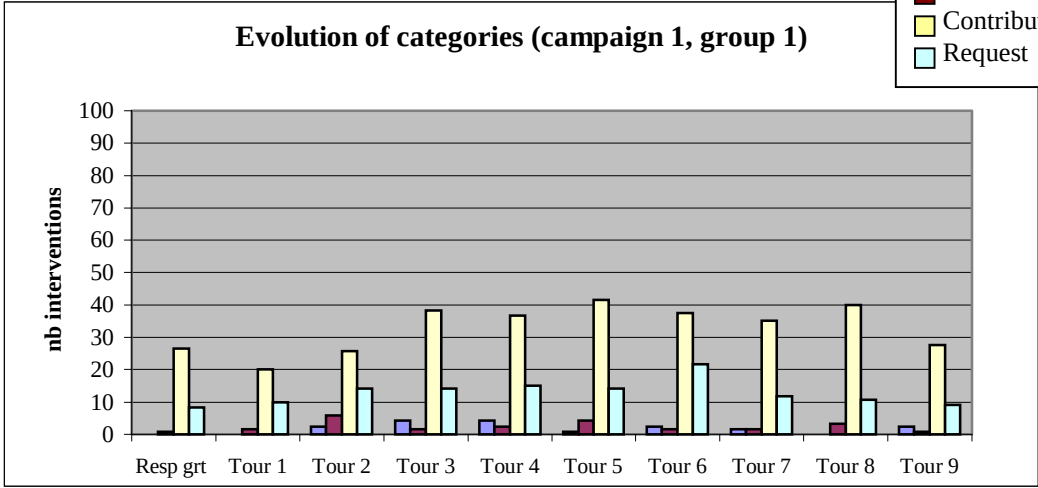
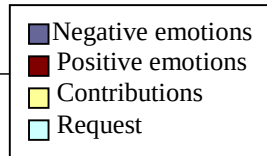


Figure 10 : Evolution of categories with players of the first session (Interactions=f(categories))

Maximum of tensions
⇒ the game course has to be stopped

During the debriefing, they have recognized that they were a little lost at the beginning of the game because they were not used to the rules. But they decided not to ask a lot about the rules and discover their application along the course of the game.

As we have already said the Bales' method has shown some information about players' learning of the game. To go further in the reflection, we think that the ignorance of the rules in the game is a metaphor of the farmers' lack of knowledge of the irrigated management when first irrigated schemes were built in the 70's. But, little by little, they learned how to manage water in these artificial areas.

The invitation of the reality in the game appears also in the game about the application of water distribution rules. At the beginning of the game, each group decides the way they want to allocate water to the plots of their watercourse. But we have seen that the rule was not systematically applied. For us, it seemed that during the game, another rule has emerged : the most important was to preserve plot sowed from stress to avoid decreases of yields. The water distribution rule, less important, was adapted to the state of the crop. We asked players in the debriefing why the distribution rule was not strictly applied? They confirm that in the real scheme the second rule is more important than the previous one and they presented different cases in which the distribution rule is temporarily modified.

At the beginning of the game they choose to draw location of plot at random. There is no rule about the allocation of plots in the game but all groups did the same. Doing so, they represented the way plots were officially distributed by donors or development agencies. The official reality⁵ was reproduced here.

The schematic representation of the world given by the game was sufficient enough to replicate some deviant behaviors: the theft of water. They recognized that thefts exist in the real world and they explained to us that several types exist. The worst is when it provoked damages to others crops.

In the debriefing, they also expressed that some rules were not accorded to reality. For example, the abacus used to compute water level on each tour (figure 3) always favors upper plots. But in the reality they found a solution to this physical rule and favor, if they want, downer plots : they play on the diameter of pipes by closing them partially with clay. This reflection about the gap between play and reality is very important because it reveals to us that the game is enough closed to reality to make farmers express they feel unease with some rules.

The most visible introduction of the reality in the game appeared when a conflict occurred in the second group at the fourth tour of the second campaign. At this time the game could not be played longer. The players involved in the conflict (participants 13 and 15) decided to ask the help of some others players. A comity was set up in which the facilitator (participant 4), intervened in the problem of the allocation of rural credit, was called and took a big part in the resolution of the conflict (figure 11). Participant 5 was the other facilitator. He talked less than "4" but he pointed out crucial elements to find the source of the conflict. The way to solve trouble in the real world was very helpful to solve the problem in the game.

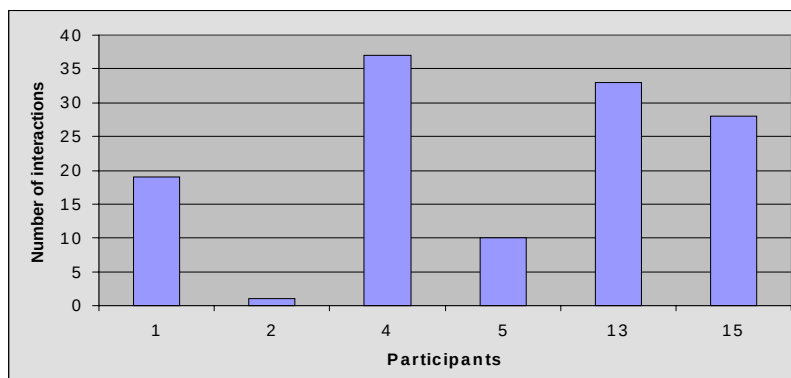


Figure 11 : Negotiation meeting to solve the conflict (Interactions = f (Participants))
("1" is the animator of the game and "2" his translator. Only personal sentences of the translator are coded.)

⁵ Cf. 4

A lot of elements enlighten the hypothesis done. For farmers, the game reproduces a lot of elements they face in the management of water. But all participants do not act the same way in the game. For some, the game was regarded as a simulation of the reality (as some said) whereas for others it was more a theater, a scene outside the real life. Therefore, some have tried to follow the real technical recommendations on crop conduct while others have just done the minimum to water, saw and harvest their crop. One seemed to use the game as a social space to show to others players that he is a good farmer in the reality (he preferred give real money rather than virtual one, to give alms as some asked him).

4. Perspectives

What is the part of social reality in players' interactions to solve a problem occurred in the game? This issue is very important to better understand what is really happening in a participatory process where RPG is used as a group decision support tool. To take a first step to tackle the link between play and reality, we proposed in this paper to test one hypothesis -the game is accepted as a schematic representation of reality- among three. We carried out a methodology to grasp what is really represented for the farmers in this game. Our methodology is based on an analysis of the interactions of people committed in negotiation processes.

The methodology presented in this paper is founded on different synergies to tackle our first hypothesis. First, the materials- sheets, video and sociological interviews - to register data in the reality observed in real negotiation processes or in the RPG are complementary. The synergy of the three is necessary to record information. Second, quantitative method - Bales' grid - and qualitative one - thematic analysis, collective or individual debriefings - are useful to analyze some aspects of the discussion of members in small groups. The synergy of these methods is a main feature of the methodology. Last, there is also a synergy between the two contexts to tackle the issue. The analysis of the RPG gives information about processes of negotiation during the game. The analysis of real negotiation processes have given data to improve our understanding of the common social background and the way it is involved in the behaviors of players. Therefore, the data grasps in one situation are helpful to understand what is going on in the second.

As we were expected, Bales' grid is a relevant method to photography situations where people are interacting. It is interesting to compare situation one to another. It gives a quantitative analysis. It shows that people are focused on the task first. It reveals that people manage tensions when they are too important to continue the task. But it is a partial method based specially on communications between participants. It reveals part of the relationships between players (through the dynamic of sentences exchanged between players), behaviors and nature of some players (facilitator, agitator, leader, the passiveness of some in the conflict). So even if it is a partial method, Bales' grid provide a lot of elements to better understand what is represented for the farmers in the RPG. On another hand, because it is a partial method, it is necessary to carry out others methods capable of giving qualitative data to complete quantitative ones.

Qualitative data were obtained thanks to sociological interviews in real negotiation process about rural credits. The analysis of the management of rural credits in the last raining season has shown stakeholders and their power to legitimate one's action, the succession of formal and informal meetings to disseminate ideas, the importance of a final formal meeting to ratify the solution proposed by all villagers. In the game, people in conflict called for a formal meeting. They asked the help of the real facilitator who showed his ability to solve problem. Debriefings were also helpful to analyze how players accept the rules and the roles. In general, they expressed the links they made with their real rules and roles. They accept the roles proposed but some rules were too restricting to make them play their roles.

The game seems to be largely accepted as a representation of the reality of irrigated scheme. The schematization done is obvious but as the farmers play the game, we assist to such an explosion of possible combinations of rules and roles, information given, and interactions between players, observers and farmers that the complexity of the real world reappears. An example of the explosion of the complexity in the game can be given through the introduction of what we called an official reality. All along this paper we were focused on reality without distinguishing an official from an unofficial one. We wonder why did players need to refer to an official reality? Did they really distinguish both levels of reality? Or does it mean that they have integrated an external rule into their system? To give answers to these questions it is necessary to go further in our research by analyzing the way reality is invited to the game. The methodology has already give some clues but it seems important to test specifically our two others hypotheses.

This distinction also asks questions about the status of the RPG as a tool in a decision process. In a longer term, a methodology is expected to analyze the influence of these tools on local processes of negotiation.

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